

DAM INSPECTION COMMENTARY GENERATION

Using Vision-Language Models

Phase 0 — Model Selection Report

Zero-Shot Evaluation of Candidate VLMs for Domain-Specific Fine-Tuning

Internship Research Project

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1. Introduction

This report documents the findings of Phase 0 of a research project aimed at developing an automated dam inspection commentary generation system. The system is intended to assist licensed civil engineers by producing structured preliminary inspection reports from dam imagery, which engineers then review and validate.

The long-term objective is to fine-tune a Vision-Language Model (VLM) on a proprietary dataset of dam images paired with professional engineering commentary, provided by the partner company. Phase 0 addresses a critical prerequisite to fine-tuning: the selection of the most suitable base model architecture through rigorous zero-shot evaluation.

2. Problem Statement

2.1 The Domain Gap Problem

General-purpose VLMs are trained on large-scale web datasets containing diverse visual content. While this yields strong performance on common visual question-answering (VQA) tasks, it produces inadequate outputs for specialized engineering inspection tasks. When prompted to describe a dam image, such models generate generic visual descriptions rather than structured technical assessments.

This was empirically confirmed during preliminary testing. Prior candidate models (PaliGemma-3B-mix, Moondream2) produced outputs such as:

Observed zero-shot output (general-purpose VQA model)

"The Hoover Dam is located at approximately 10 miles north of Las Vegas, Nevada."

This output demonstrates two fundamental failures: (1) the model retrieved memorized factual information rather than performing visual analysis, and (2) the output bears no resemblance to professional engineering inspection language.

2.2 Why Prior Models Were Eliminated

The initial candidate pool included PaliGemma-3B-mix and Moondream2. Both were eliminated for the following reasons:

- PaliGemma-3B-mix was trained on short-answer VQA tasks and is architecturally constrained to produce 1-2 sentence outputs. No prompt engineering can overcome this output length limitation.
- Moondream2 exhibited the same VQA-optimized architecture and produced similarly shallow outputs unsuitable for multi-section inspection reports.
- Both models defaulted to landmark recognition and factual recall rather than visual structural analysis when presented with famous dam imagery.

These findings motivated a complete revision of the candidate pool to focus exclusively on instruction-tuned VLMs capable of extended structured text generation.

3. Methodology

3.1 Candidate Model Selection Criteria

Three replacement candidates were selected based on the following criteria:

- Instruction-tuned variant available — model must be fine-tuned for instruction following, not merely text completion.
- Capable of extended structured output — model must generate multi-paragraph, multi-section responses.
- Compatible with 4-bit / 8-bit quantization — must fit within 16GB VRAM of a T4 GPU (Google Colab free tier).
- Open-source with HuggingFace PEFT support — required for the QLoRA fine-tuning pipeline in Phase 1.

The three selected candidates are summarized below:

Model	Params	Provider	Key Characteristic
LLaVA-1.5 7B	7B	LLaVA team / HuggingFace	Established instruction-tuned VLM; strong baseline for long-form generation
Qwen2-VL 2B Instruct	2B	Alibaba (2024)	Dynamic resolution mechanism; strong visual grounding; very memory-efficient
InternVL2 2B	2B	OpenGVLab	Competitive on dense description benchmarks; strong vision-language alignment

3.2 Proxy Dataset

In the absence of the company dataset at this stage, a proxy dataset of N=9 publicly available dam images was assembled from open-access sources. The dataset was curated to cover the principal dam typologies encountered in professional inspection practice: concrete gravity, arch, and embankment configurations, as well as images capturing operational spillway conditions.

This proxy dataset serves exclusively as a zero-shot evaluation benchmark. It will be replaced in its entirety by the company dataset in Phase 1, at which point all images and annotations will originate from real field inspection records.

3.3 Prompt Engineering

A single standardized prompt was designed and applied identically across all three candidate models to ensure a fair comparison. The prompt follows the role-task-format-uncertainty design pattern:

- Role: establishes the model as a licensed civil engineer specializing in dam safety.
- Task: requests description of seven specific structural and hydraulic aspects using imperative sentences rather than numbered headings to avoid document-template completion behavior.
- Format: specifies formal engineering language and visual specificity.
- Uncertainty protocol: instructs the model to explicitly state when features are not assessable from the image rather than hallucinating observations.

The prompt was deliberately framed as a visual description task rather than a report-writing task. Earlier experiments demonstrated that prompts containing the phrase 'write a report' caused models to generate blank document templates (e.g., "Dam Inspection Report — [Date] ---") rather than substantive content. The final prompt begins with 'describe what you observe' to anchor the model in visual analysis mode.

3.4 Generation Parameters

The following generation parameters were applied uniformly across all models:

- max_new_tokens = 600 — sufficient for a complete 7-section inspection commentary.
- do_sample = True, temperature = 0.3 — low-temperature sampling to prevent template-locking while maintaining output consistency.
- top_p = 0.9 — nucleus sampling to avoid degenerate repetition.
- repetition_penalty = 1.3 — explicit penalty against repeating token sequences.
- no_repeat_ngram_size = 4 — prevents any 4-gram from appearing more than once in the output.

A forced generation prefix ('Upon visual inspection of this dam, I observe the following:') was prepended to the assistant turn to bypass document-header generation behavior observed in earlier experiments.

3.5 Evaluation Metrics

Four proxy metrics were computed automatically for each model across all N=9 images:

- Average Word Count — measures output verbosity. Higher values indicate more complete coverage.
- Average Section Coverage — fraction of the 7 required inspection sections present, detected via keyword group matching. Values range from 0 (no sections) to 1.0 (all sections present).
- Average Technical Term Density — count of engineering domain vocabulary terms per 100 output words. A curated lexicon of 45 terms was used, spanning dam typology, structural components, observed defects, and inspection methodology vocabulary.
- Average Inference Time — mean per-image generation time in seconds on a T4 GPU. Relevant for deployment feasibility assessment.

These metrics are explicitly proxy measures. They assess structural and linguistic output characteristics, not engineering correctness. A full domain-specific human evaluation rubric will be applied in Phase 2 following fine-tuning.

4. Results

4.1 Quantitative Comparison

The table below summarizes the quantitative evaluation results across all three candidate models:

Metric	LLaVA-1.5 7B	Qwen2-VL 2B	InternVL2 2B	Best
Avg. Word Count	114.7	415.9	288.0	Qwen2-VL
Avg. Inference Time (s)	13.36	48.79	41.58	LLaVA-1.5
Avg. Section Coverage	0.428	0.968	1.000	InternVL2
Avg. Technical Density	2.78	2.82	4.85	InternVL2
Sample Count (N)	9	9	9	—

4.2 Analysis

LLaVA-1.5 7B

Despite having the largest parameter count (7B) among the three candidates, LLaVA-1.5 produced the shortest outputs (114.7 words average) with the lowest section coverage (0.428). This indicates that fewer than half of the required inspection sections were present on average — insufficient for a structured engineering report. LLaVA-1.5 also exhibited the fastest inference time (13.36s), but output quality is the primary criterion for this task. This model is eliminated from further consideration.

Key finding

Model parameter count is not a reliable predictor of output suitability for structured domain-specific generation tasks. LLaVA-1.5 7B was outperformed on all quality metrics by both 2B models.

Qwen2-VL 2B Instruct

Qwen2-VL produced the highest average word count (415.9) and near-complete section coverage (0.968). Its dynamic resolution vision mechanism appears to facilitate more detailed visual observation compared to fixed-resolution models. Technical term density (2.82 per 100 words) was comparable to LLaVA-1.5, however, and inference time was the highest of the three candidates (48.79s per image on T4).

InternVL2 2B

InternVL2-2B achieved perfect section coverage (1.000) across all nine proxy images and the highest technical term density by a substantial margin (4.85 per 100 words — 72% higher than Qwen2-VL). Inference time (41.58s) was 7 seconds faster than Qwen2-VL. Average word count (288.0) was lower than Qwen2-VL but with higher domain vocabulary concentration, indicating more precise engineering language rather than verbose general description.

4.3 Qualitative Assessment

Visual inspection of InternVL2-2B outputs confirmed the quantitative findings. The model correctly identified dam typology (concrete gravity), enumerated relevant structural components (crest, downstream face, spillway, abutments), and generated plausible maintenance recommendations including geotechnical surveys and integrity assessments.

Two categories of limitations were identified in the zero-shot outputs:

- **Hallucinated defect observations:** The model generated plausible-sounding surface condition statements (e.g., 'cracks along certain sections') that cannot be verified from the image alone, indicating reliance on prior knowledge rather than visual grounding for this specific output section.

- Occasional component misidentification: Structural terms were sometimes applied with imprecise definitions (e.g., conflating downstream face description with abutment function).

These failure modes are consistent with known limitations of general-purpose VLMs on specialized inspection tasks and constitute the primary technical motivation for domain-specific fine-tuning. They are expected to be substantially reduced after training on annotated dam inspection imagery.

5. Model Selection Decision

Selected model for Phase 1 fine-tuning: InternVL2-2B (OpenGVLab/InternVL2-2B)

InternVL2-2B is selected based on superior section coverage (1.000), highest technical term density (4.85 per 100 words), and acceptable inference time on T4 hardware. It demonstrates the strongest domain alignment of the three candidates under zero-shot conditions, providing the most suitable base for domain-specific fine-tuning.

The selection is justified on the following grounds:

- **Perfect (1.000 across N=9 images). All seven required inspection sections were present in every output, establishing a complete structural foundation that fine-tuning can then populate with domain-accurate content.:** Section coverage
- **4.85 per 100 words — 72% higher than the next best candidate (Qwen2-VL: 2.82). This indicates stronger intrinsic alignment with engineering vocabulary prior to any domain adaptation.:** Technical term density
- **41.58s per image on T4 — 7 seconds faster than Qwen2-VL at comparable output length, relevant for future deployment in an engineering workflow context.:** Inference efficiency
- **InternVL2 is fully compatible with HuggingFace PEFT and supports QLoRA fine-tuning, which is the planned approach for Phase 1 given the T4 hardware constraint.:** Fine-tuning ecosystem

6. Limitations and Future Work

The following limitations of this Phase 0 study should be noted:

- Proxy dataset: The N=9 evaluation images are publicly available photographs of well-known dams, not field inspection imagery. Results may differ on the company dataset, which will feature different image quality, angles, and dam scales.

- Zero-shot conditions only: No few-shot examples or domain-specific context were provided. Phase 1 fine-tuning is expected to substantially improve all metrics.
- Automatic metrics only: Section coverage and technical density are structural proxies. Engineering correctness was not assessed at this stage. A human expert evaluation rubric will be introduced in Phase 2.
- Single prompt variant: Only one prompt formulation was evaluated. Prompt sensitivity analysis across multiple formulations may yield different relative rankings.

Phase 1 will address these limitations by:

- Fine-tuning InternVL2-2B on the company dataset using QLoRA on a T4 GPU.
- Introducing a structured human evaluation rubric scored by domain experts.
- Conducting an ablation study on training data volume (e.g., 50 vs. 100 vs. 200 image-commentary pairs) to quantify the data efficiency of the fine-tuning approach.

7. Conclusion

Phase 0 established through systematic zero-shot evaluation that InternVL2-2B is the most suitable base architecture for domain-specific fine-tuning on dam inspection commentary generation. The evaluation demonstrated that model parameter count is not a reliable proxy for task suitability — LLaVA-1.5 7B was outperformed on all quality metrics by both 2B models. InternVL2-2B's superior section coverage and technical vocabulary density under zero-shot conditions provide a strong foundation for the fine-tuning phase. The documented failure modes — hallucinated defect observations and occasional component misidentification — represent well-understood VLM limitations that domain-specific training is directly designed to address.